

ANALYZING POPULATION TRENDS AND PREDICTING COUNTY-LEVEL POPULATION GROWTH IN KENYA USING 2019 KNBS CENSUS DATA

A Data Science Project — Year 1, Semester 1

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Abstract

This project analyzes population distribution across Kenya's 47 counties using data from the 2019 Kenya Population and Housing Census (KNBS). It identifies key trends in population size, gender distribution, and population density, and applies a simple exponential growth model — based on the national inter-censal growth rate of 2.2% — to project county populations to 2029. The findings show that Nairobi City, Kiambu, and Nakuru are the most populous counties, while Nairobi, Mombasa, and Kisumu exhibit the highest population densities. The results offer a basic, data-driven basis for discussions on resource allocation and infrastructure planning at the county level.

1. Introduction

Kenya conducts a national census every ten years to inform planning, budgeting, and resource allocation across its 47 counties. The 2019 Kenya Population and Housing Census (KPHC), conducted by the Kenya National Bureau of Statistics (KNBS), recorded a national population of 47,564,296 people. Understanding how this population is distributed — and how it is likely to grow — is essential for policymakers, county governments, and planners.

This project uses the officially published county-level census data to explore population patterns and produce a simple forward projection for 2029.

1.1 Objectives

- To analyze the distribution of Kenya's population by county.
- To identify the most populous and most densely populated counties.
- To examine the relationship between total population and population density.
- To project 2029 county populations using the national inter-censal growth rate (which give out the prediction of 2029 census)

1.2 Scope

This analysis covers all 47 counties using the 2019 census figures for population, sex distribution, land area, and population density, as published in Volume I of the 2019 KPHC report.

2. Data Source and Methodology

Data was sourced directly from the KNBS publication “2019 Kenya Population and Housing Census, Volume I: Population by County and Sub-County” (November 2019). The relevant tables (Table 2.2 and Table 2.4) provided population by sex, land area, and population density for each county.

(The data was compiled into a structured CSV file and analyzed using Python, with the pandas library for data handling and matplotlib for visualization. Google Colab was used as the development environment)

3. Data Cleaning

The source data was already validated and published by KNBS, so minimal cleaning was required. Column names were standardized (e.g., removing spaces, using consistent capitalization), and the data was checked to confirm all 47 counties were present with no missing values.

4. Exploratory Data Analysis(EDA)

4.1 Most Populous Counties

Figure 1 shows the ten most populous counties in Kenya as of 2019. Nairobi City leads by a wide margin with over 4.3 million people, followed by Kiambu and Nakuru. This concentration reflects the pull of urban and peri-urban areas for employment and services which give out clear prediction 2029 census.

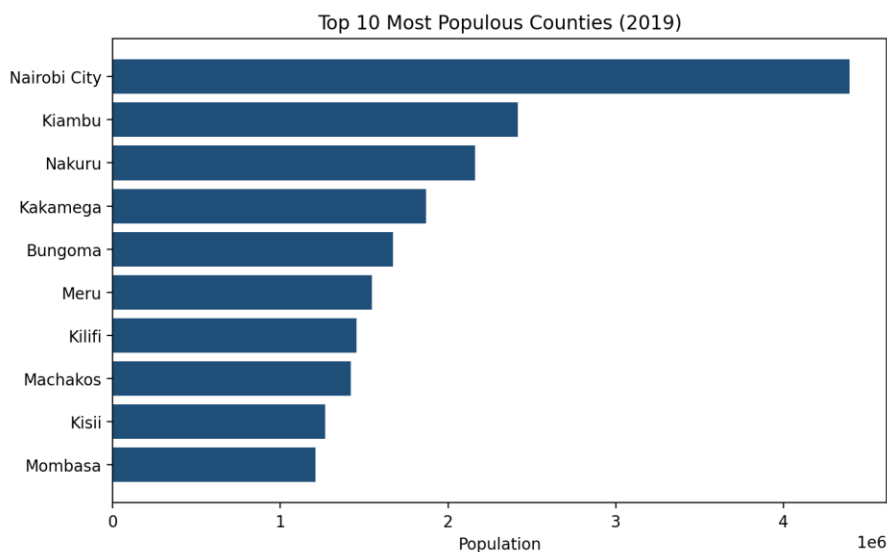


Figure 1: Top 10 Most Populous Counties (2019)

4.2 Population Density

Figure 2 shows the ten most densely populated counties. Unlike the population ranking, this list is dominated by smaller, urbanized counties such as Nairobi City, Mombasa, and Kisumu, as well as densely settled agricultural counties like Kisii and Vihiga, which have small land areas relative to their populations.

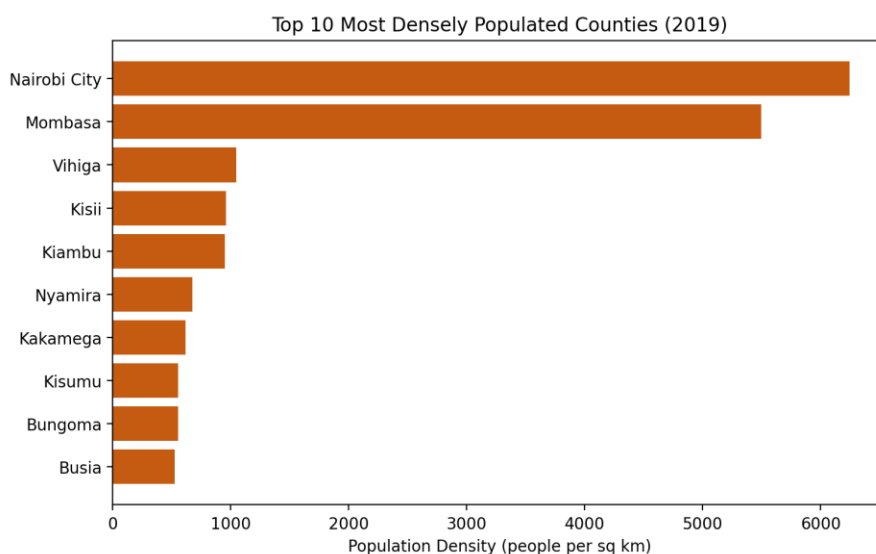


Figure 2: Top 10 Most Densely Populated Counties (2019)

4.3 Population vs. Density

Figure 3 plots total population against population density for all 47 counties. Most counties cluster in the lower-left region, indicating relatively low population and low density. Nairobi City stands out as an outlier

with both high population and extreme density, while several smaller counties (labeled) show very high density despite modest total population — indicating limited land area rather than large population size.

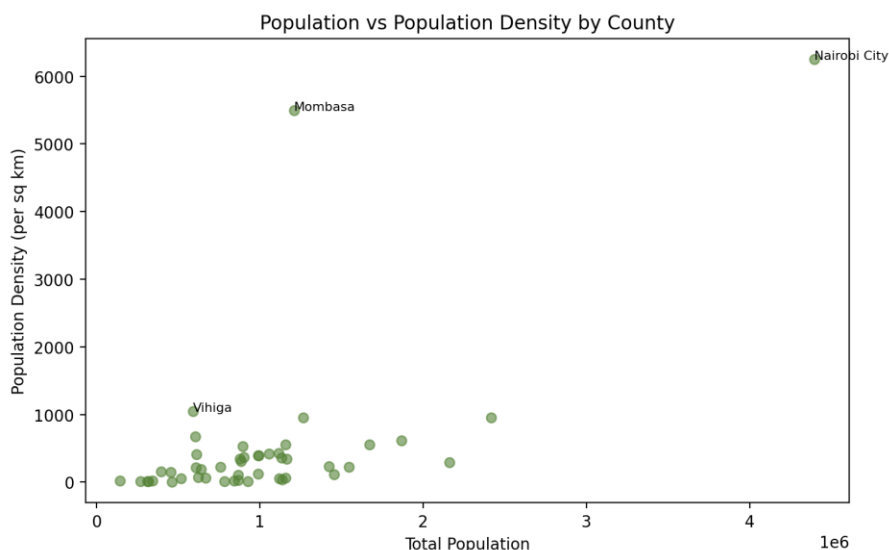


Figure 3: Population vs Population Density by County

5. Predictive Model: 2029 Population Projection

To estimate future population, a compound growth model was applied using the national inter-censal growth rate of 2.2% per year, as reported by KNBS for the 2009–2019 period:

$$\text{Predicted Population (2029)} = \text{Population (2019)} \times (1 + 0.022)^{10}$$

Table 1 shows the projected 2029 population for the ten currently most populous counties.

County	2019 Population	Predicted 2029 Population
Nairobi City	4,397,073	5,466,038
Kiambu	2,417,735	3,005,506
Nakuru	2,162,202	2,687,851
Kakamega	1,867,579	2,321,603
Bungoma	1,670,570	2,076,699
Meru	1,545,714	1,921,490
Kilifi	1,453,787	1,807,215
Machakos	1,421,932	1,767,615
Kisii	1,266,860	1,574,844
Mombasa	1,208,333	1,502,089

Table 1: Predicted 2029 Population for Top 10 Counties

6. Discussion

- Nairobi City, Kiambu, and Nakuru remain Kenya's most populous counties and are projected to continue leading in absolute population by 2029.
- Urban and peri-urban counties show the highest population density, while arid and semi-arid counties (e.g., Marsabit, Turkana) have vast land area but low density.

- The 2.2% national growth rate is a simplification — real growth rates vary significantly by county due to migration, fertility, and urbanization patterns. Fast-growing counties like Kiambu and Nakuru may be underestimated, while some rural counties may be overestimated.
- This limitation means the projections should be treated as an approximate national-average scenario, not a precise county-specific forecast.

7. Conclusion and Recommendations

This project demonstrates a basic but complete data science workflow — from raw government data to cleaned dataset, exploratory analysis, and a simple predictive model — applied to Kenya's 2019 census data. The findings confirm that population growth in Kenya is highly uneven across counties, with urban centers absorbing a disproportionate share of national population growth.

For more accurate forecasting, future work should incorporate county-specific historical growth rates (comparing 2009 and 2019 figures directly) and consider non-linear growth models such as ARIMA or logistic growth curves, which better capture how growth rates change over time.

These findings could support county governments and national planners in prioritizing infrastructure, healthcare, and education investment in fast-growing counties.

References

Kenya National Bureau of Statistics (KNBS). (2019). 2019 Kenya Population and Housing Census, Volume I: Population by County and Sub-County. Nairobi: KNBS.